

Chapter 4

Data Link Layer

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- LLC and MAC sub layer overview
- Flow control – stop and wait, sliding window
- Error control – Automatic Repeat Request (ARQ), stop and wait ARQ, Sliding window ARQ
- Error control mechanism
 - Error detection: parity check, checksum, CRC
 - Error correction: hamming code
- ALOHA, CSMA/CD
- Ethernet, Fast Ethernet, Gigabit Ethernet
- Virtual Circuit Switching: Frame Relay, ATM, X.25

Data Link Layer

Network Layer		
Data link Layer	LLC sublayer	IEEE 802.15 Blue tooth
	MAC sublayer	IEEE 802.11 WiFi
Physical Layer		IEEE 802.15 Ethernet

The services provided by the data link layer are:

1) **Flow control**

- ✓ An issue that occurs at every levels is how to keep a fast sender from swapping a slow receiver with data. E.g. some kind of feedback from the receiver to the sender

2) **Framing**

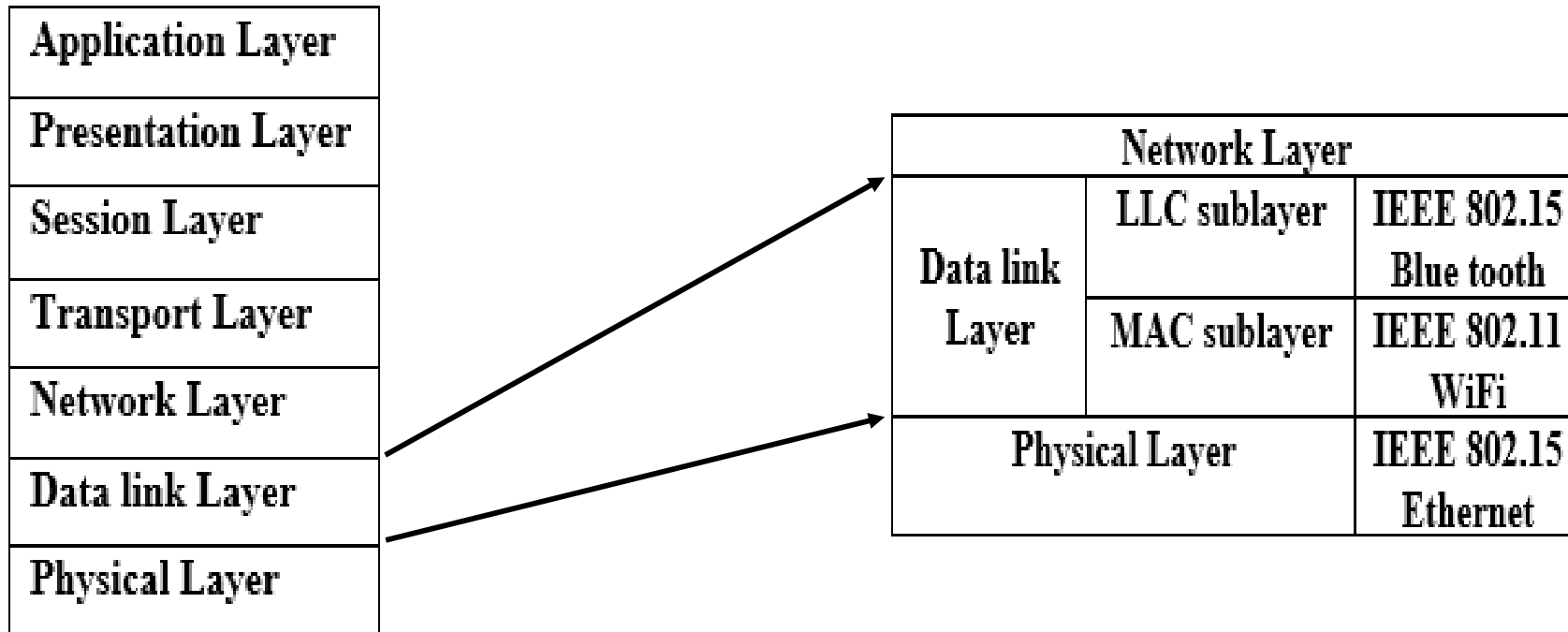
- ✓ Breaking the bits stream up into frames is know as framing

3) **Physical Addressing / MAC addressing**

- Physical addressing and MAC addressing are both related to the MAC sublayer of the data link layer, and are concerned with identifying devices on a network.
- Also known as MAC addressing, this is the process of assigning a unique identifier to a network interface controller (NIC) on a device. This identifier is used to communicate on a network segment.

Data Link Layer

- Data Link Layer resides above the physical layer and below the network layer.
- The function of data link layer is to accept the packets from the network layer and convert it into frame



Data Link Layer

Network Layer		
Data link Layer	LLC sublayer	IEEE 802.15 Blue tooth
	MAC sublayer	IEEE 802.11 WiFi
Physical Layer		IEEE 802.15 Ethernet

4) Error Control

- ✓ Error control is an important issue because physical communication circuits are not perfect. Many errors detecting codes are know, but both ends of the connection must agree on which one is being used.

MAC address

A MAC address is a unique identifier for a device on a local network. It's also known as a physical address, and is usually burned into the NIC's read-only memory (ROM) chip.

Data Link Layer

Network Layer		
Data link Layer	LLC sublayer	IEEE 802.15 Blue tooth
	MAC sublayer	IEEE 802.11 WiFi
Physical Layer		IEEE 802.15 Ethernet

- Datalink sublayer constitutes of Logical Link Control and Media Access Control
- Logical Link Control (LLC)sublayer
 - Also known as data link Control (DLC) sublayer
 - Handles communication between upper and lower layer
- Media Access Control (MAC)sublayer
 - Constitutes the lower sublayer of the data link layer
 - It is normally implemented by hardware, typically in the computer network interface card (NIC)

MAC sublayer

Network Layer		
Data link Layer	LLC sublayer	IEEE 802.15 Blue tooth
	MAC sublayer	IEEE 802.11 WiFi
Physical Layer		IEEE 802.15 Ethernet

- The two major responsibilities of MAC sublayer are
 - a) Data Encapsulation
 - ✓ MAC sub layer adds a header and trailer to the network layer PDU
 - ✓ Frame assembly before transmission and frame disassembly after/upon reception of a frame
 - b) Responsible for the placement of frames on the media and the removal of frames from the media
 - ✓ On the media → sender side
 - ✓ From the media → receiver side
 - ✓ Communicates directly with the physical layer

Protocol Data Unit (PDU)

PDU is a significant term related to the initial four layers of the OSI model

In layer 1, PDU is a bit

In layer 2, PDU is a frame

In layer 3, PDU is a packet

In layer 4, PDU is a segment

In layer 5, PDU is a data

Which sublayer is responsible for which of the services?

<u>Services</u>	<u>Sub Layer</u>
Flow Control	LLC or DLC
Framing	MAC
Physical Addressing	MAC
Error Control	MAC
Access Control	MAC

Flow Control

- It is a set of procedures that tells the sender how much data it can transmit before it must wait for an acknowledgement for the receiver
- In fact any receiving device has a limited speed at which it can process incoming data and a limited amount of memory, called a buffer, in which to store incoming data until they are processed.
- Two methods have been developed to control the flow of data across the communication links
 - 1) Stop and wait protocol
 - 2) Sliding window protocol

Flow Control

1) Stop and wait protocol

- Here the sender sends on frame and waits for an acknowledgement before sending the next frame
- This process of alternatively sending and waiting repeats till the sender transmits an end of transmission (EOT) frame

Advantages

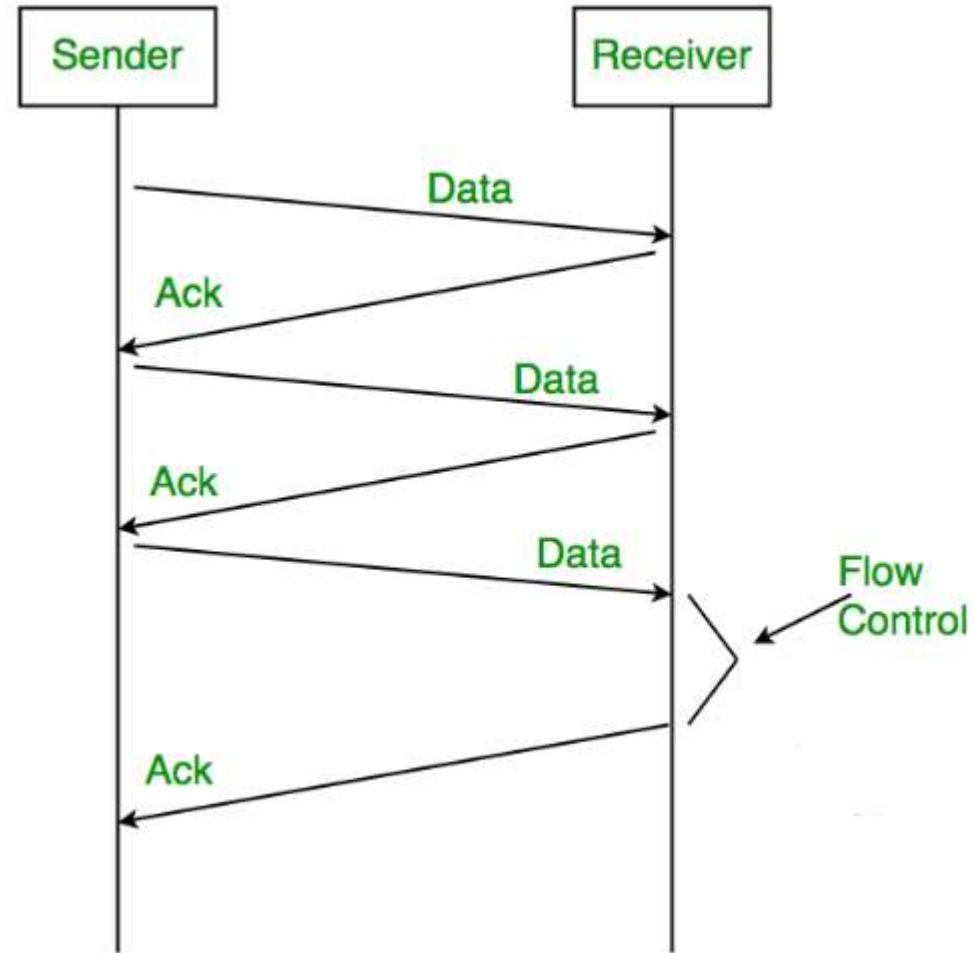
- Idea of stop and wait protocol is simple
- Each frame is checked and acknowledged before the next frame is sent

Disadvantages

- If the acknowledgement does not arrive at time, after a certain period of time, the sender times out and retransmits the original frame.
 - One frame at time
 - Poor utilization of the bandwidth
 - Poor performance of the network (Bandwidth is not fully utilized)

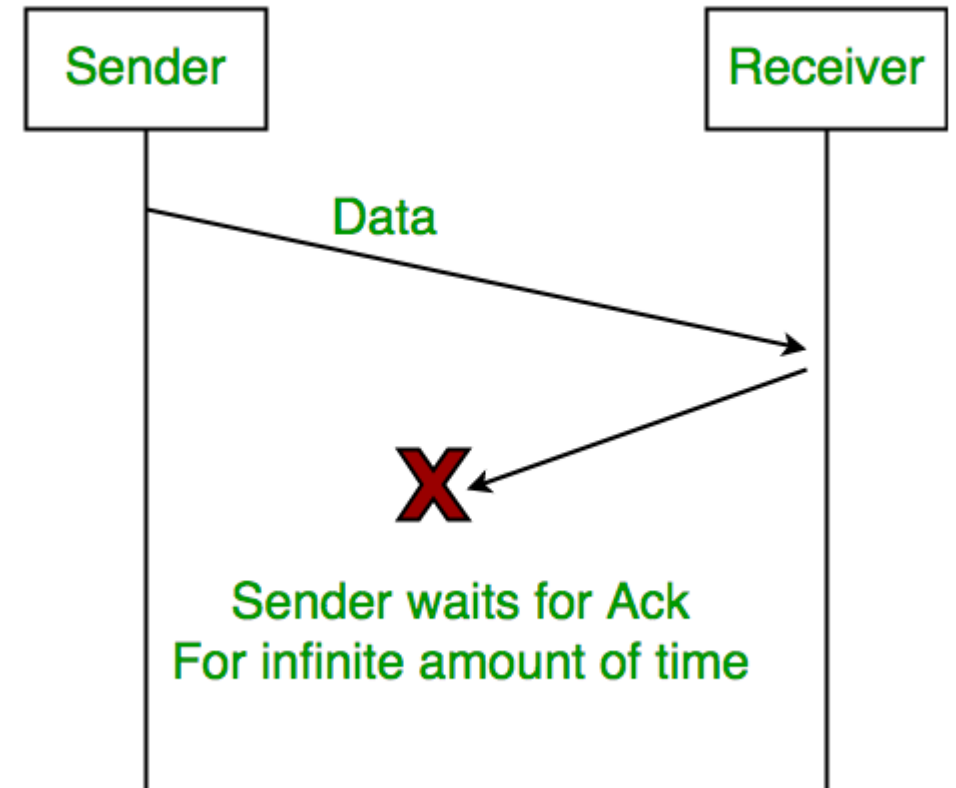
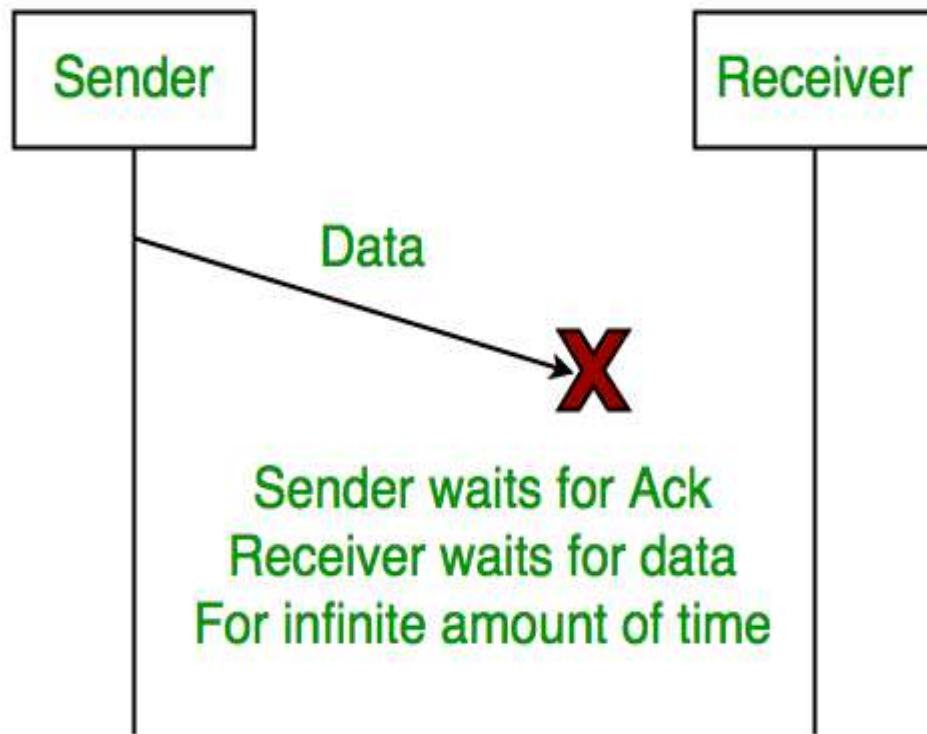
Flow Control

1) Stop and wait protocol



Flow Control

1) Stop and wait protocol



Flow Control

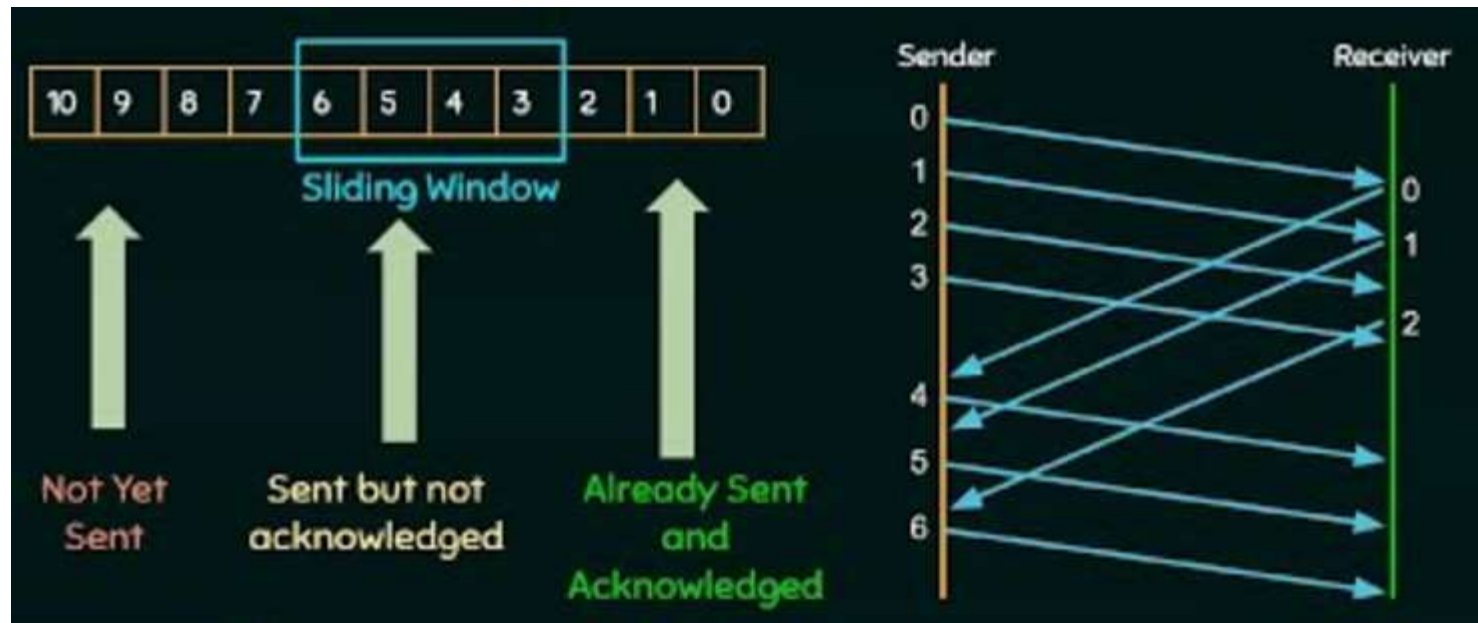
2) Sliding window protocol

- Here we can send multiple frames at a time.
- Number of frames to be sent is based on window size
- Each frame is numbered and is known as sequence number

Flow Control

2) Sliding window protocol

- Suppose we have a sender and a receiver.
- The sender has 11 frames to send
- How many frames can be sent is decided by a parameter called window size. Here suppose window size = 4



Flow Control

2) Sliding window protocol

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Here suppose window size = 4

Error Control

- Mainly it refers to error detection and retransmission
- Whenever an error is detected in an exchange, a negative acknowledgement (NAK) is returned and the specified frames are retransmitted. This process is called automatic repeat request (ARQ)
- Error control in the data link layer is based on automatic repeat request, which means retransmission of data in three cases:
 - i) damaged data frame
 - ii) lost data frames and
 - lii) Lost acknowledgement frame

Error Control

Types of ARQ

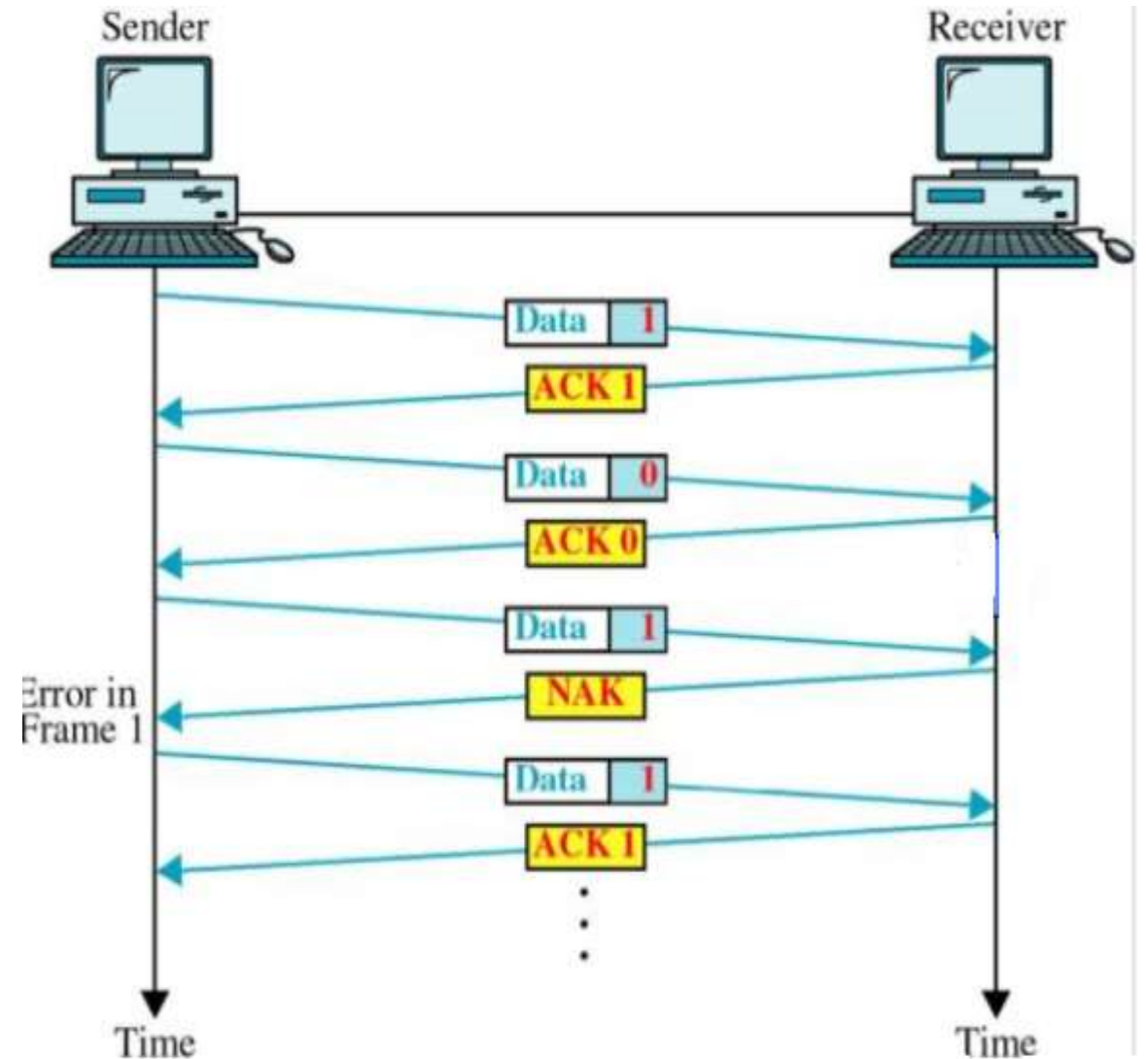
1) Stop and wait ARQ

2) Sliding window ARQ

- Selective repeat ARQ
- Go back n ARQ

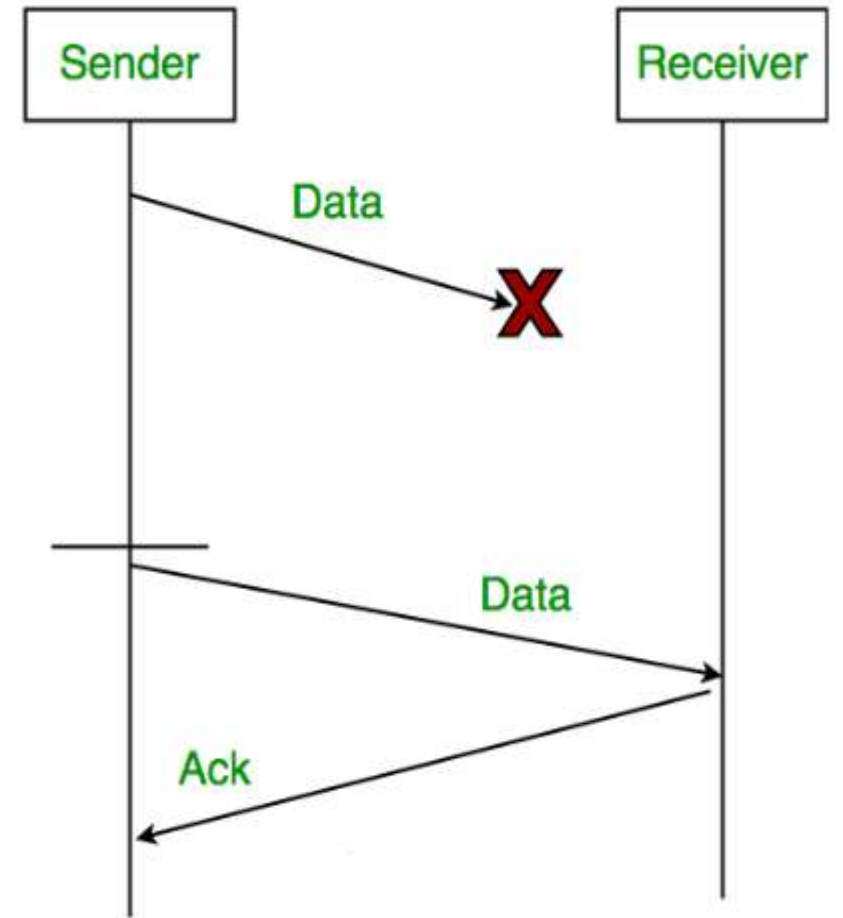
Stop and wait ARQ – damaged data frames

- When a frame is discovered by the receiver to contain an error, it returns a NAK frame and the sender retransmits the signal.



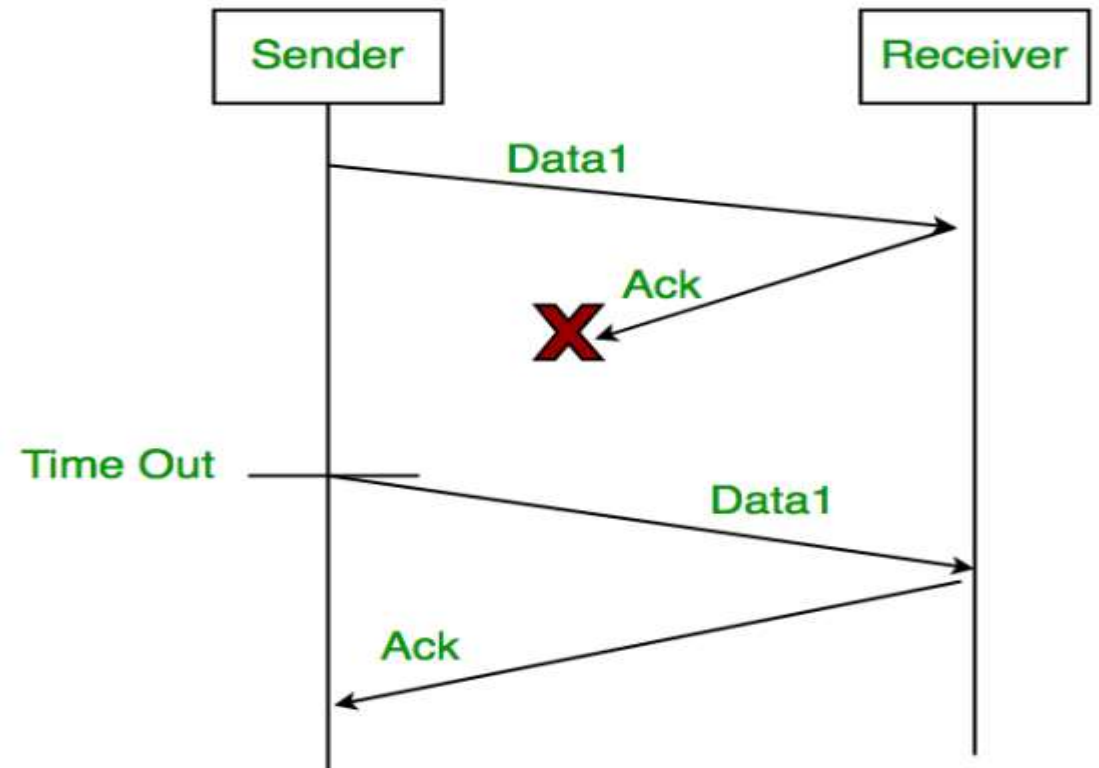
Stop and wait ARQ – lost data frames

- Here sender is equipped with a timer and it starts every time a data frame is transmitted
- If an acknowledgement has not been received within the time limit, the sender assumes that the last data frame was lost in transit and sends it again.



Stop and wait ARQ – lost acknowledgement frames

- Here sender is equipped with a timer and it starts every time a data frame is transmitted
- If an acknowledgement has not been received within the time limit, the sender retransmits every frame transmitted since the last ACK



Error Control

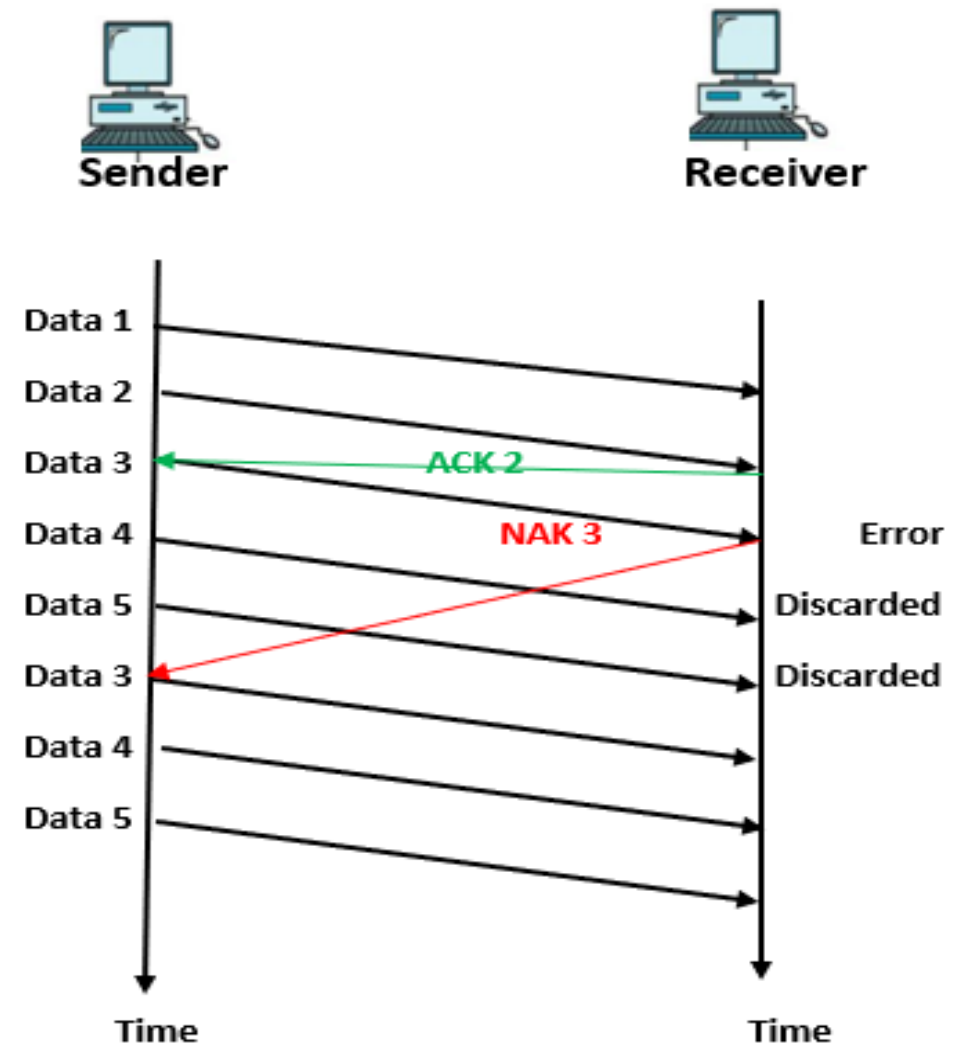
Types of ARQ

2) Sliding window ARQ

- For continuous transmission of error control, two protocols are the most popular.
- One is **go-back-n ARQ** and the other is **Selective repeat ARQ**

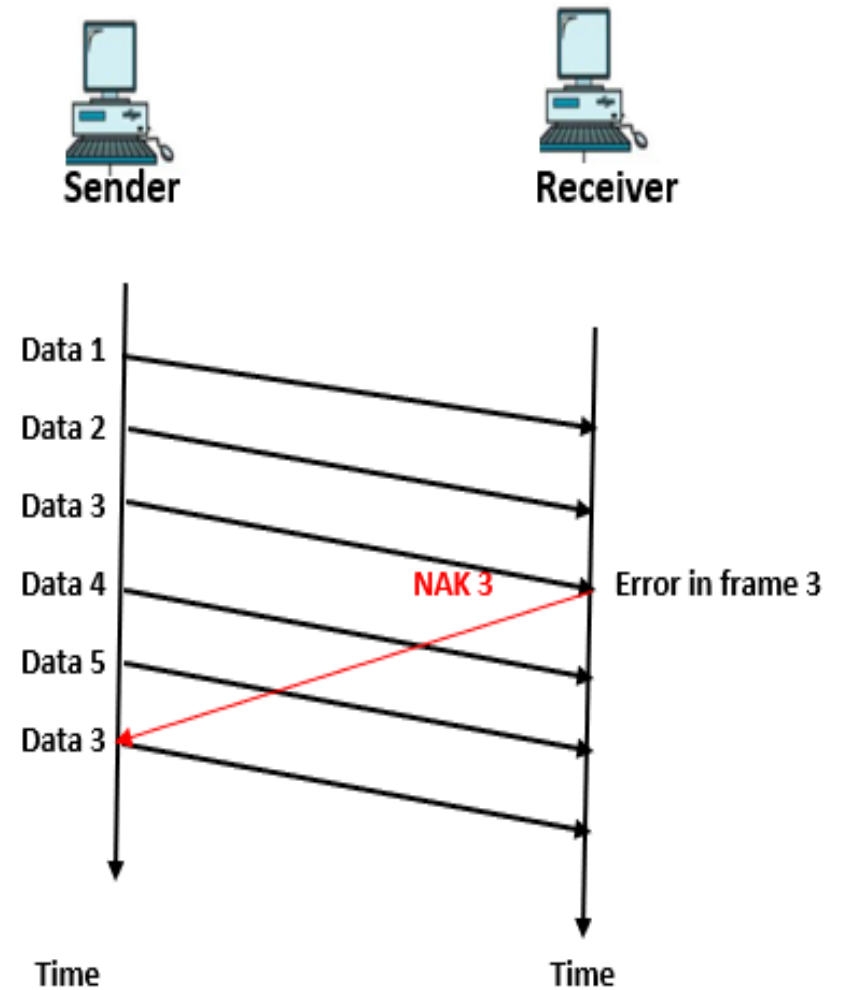
Go-Back-N ARQ

- In this method, if one frame is lost or damaged, all frames sent since the last acknowledgement are re-transmitted.
- Here, in the figure, ACK2 tells the sender that frames 1 and frame 2 have all been accepted.
- NAK3 tells that data3 is damaged, so frames 4 and frame 5 are discarded. Hereby the sender retransmits frame 3, 4, 5 and the process continues



Selective Repeat ARQ

- In selective repeat ARQ, only the specific damaged or lost frame is retransmitted. If a frame is corrupted in transit, a NAK is returned and the frame is resent out of sequence



Go-Back N ARQ versus Selective Repeat ARQ

Go-Back N ARQ

- Here out of order packets are not accepted and the entire frame since the last error is again re-transmitted
- Neither sender nor receiver needs sorting

Selective Repeat ARQ

- Here out of order packets are accepted
- The receiver side needs sorting to maintain the originality of the data

Error control mechanism

- Error detection
 - parity, checksum, CRC
- Error correction
 - hamming code

Solve

- Given data = 1100, and generator = x^3+x+1 . Implement CRC in the hardware side.
- Determine which bit is in error in the odd parity. Hamming coded character is 1100111

Channel Access Mechanisms

- ALOHA
 - Pure ALOHA
 - Slotted ALOHA
- CSMA/CD

Channel Access Mechanisms

- ALOHO
 - The earliest technique for accessing any shared transmission medium is ALOHA

Random Access

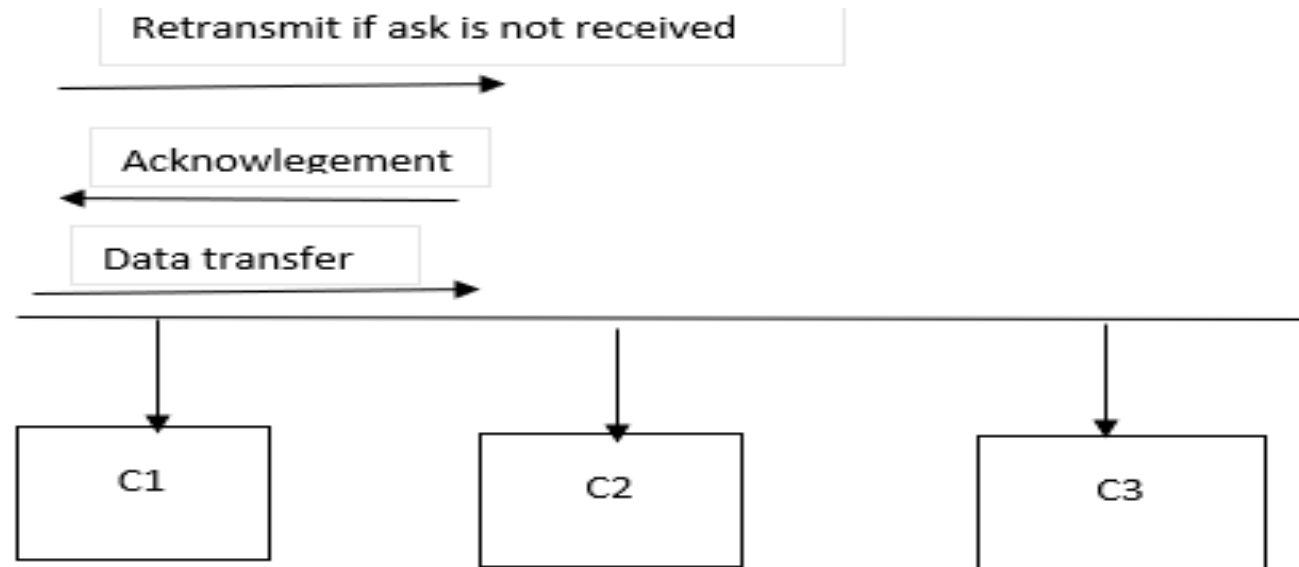
- Each station has the right to the medium without being controlled by any other station.
 - Two features gives the method its name:
 - There is no schedule time for a station to transmit: transmission is random among stations
- **Collision:** an access conflict occurs when more than one station tries to send, as a result the frame will be either destroyed or modified.

Aloha

- The earliest technique for accessing any shared transmission medium is Aloha
- Random access protocol for data transfer
- Allows subscriber to transmit whenever they have data to send
- The transmitter listens for a time interval to determine feedback from the receiver
 - If an acknowledgement is received, the transmitter knows that the data has been received successfully; otherwise the data is re-transmitted
 - If the station fails to receive an acknowledgment, after repeated transmission it gives up.

Aloha

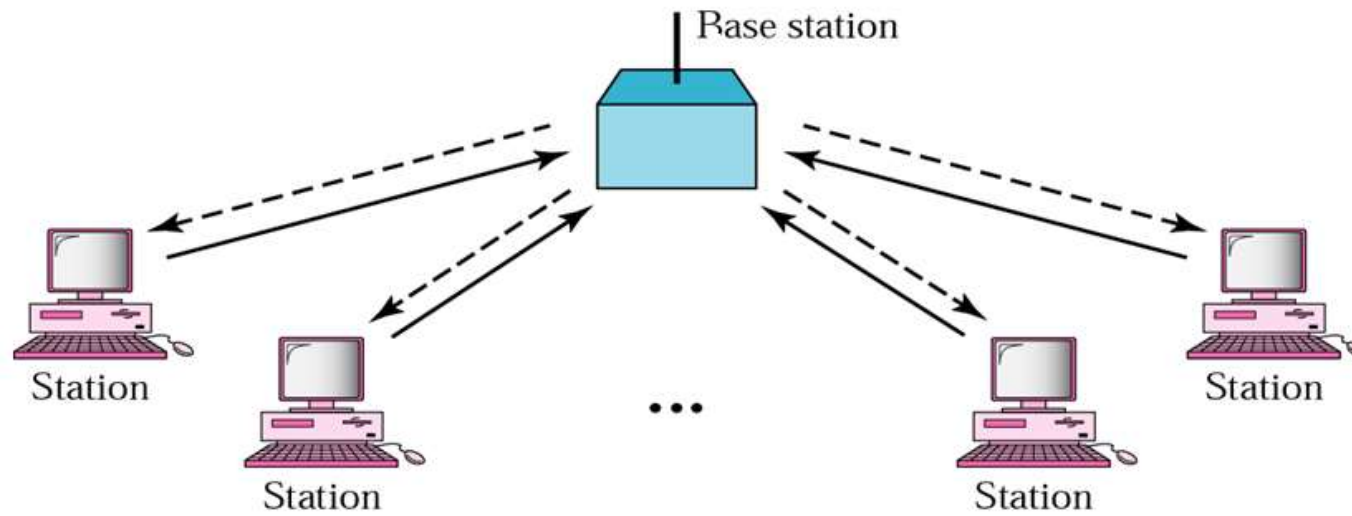
- Aloha is simple to implement but causes degradation in performance as traffic increases
- Increase load causes high number of traffic



Pure Aloha

- *The earliest random access method developed at the Univ. of Hawaii in the early 1970s*
- *Designed for a radio (wireless) LAN*
- *Simple method. Each station sends a frame whenever it has a frame to send.*
- *Since there is only one channel to share, there is the possibility of **collision** between frames from different stations.*

Pure Aloha



Slotted Aloha

- Slotted Aloha is a network protocol that improves the efficiency of the original Aloha protocol by dividing time into discrete slots.
- It is a refinement over the pure aloha. It requires that time be segmented into slots of a fixed length be exactly equal to the packet transmission time
- **How it works?**
 - In Slotted Aloha, users can only transmit data at the beginning of a time slot, which prevents partial overlap and reduces collisions.
- **Benefits**
 - Slotted Aloha increases the network's capacity and maximum throughput.
- **Uses**
 - Slotted Aloha is used in many applications, including computer networks, wireless sensor networks, GSM cellular networks, and military satellite communications

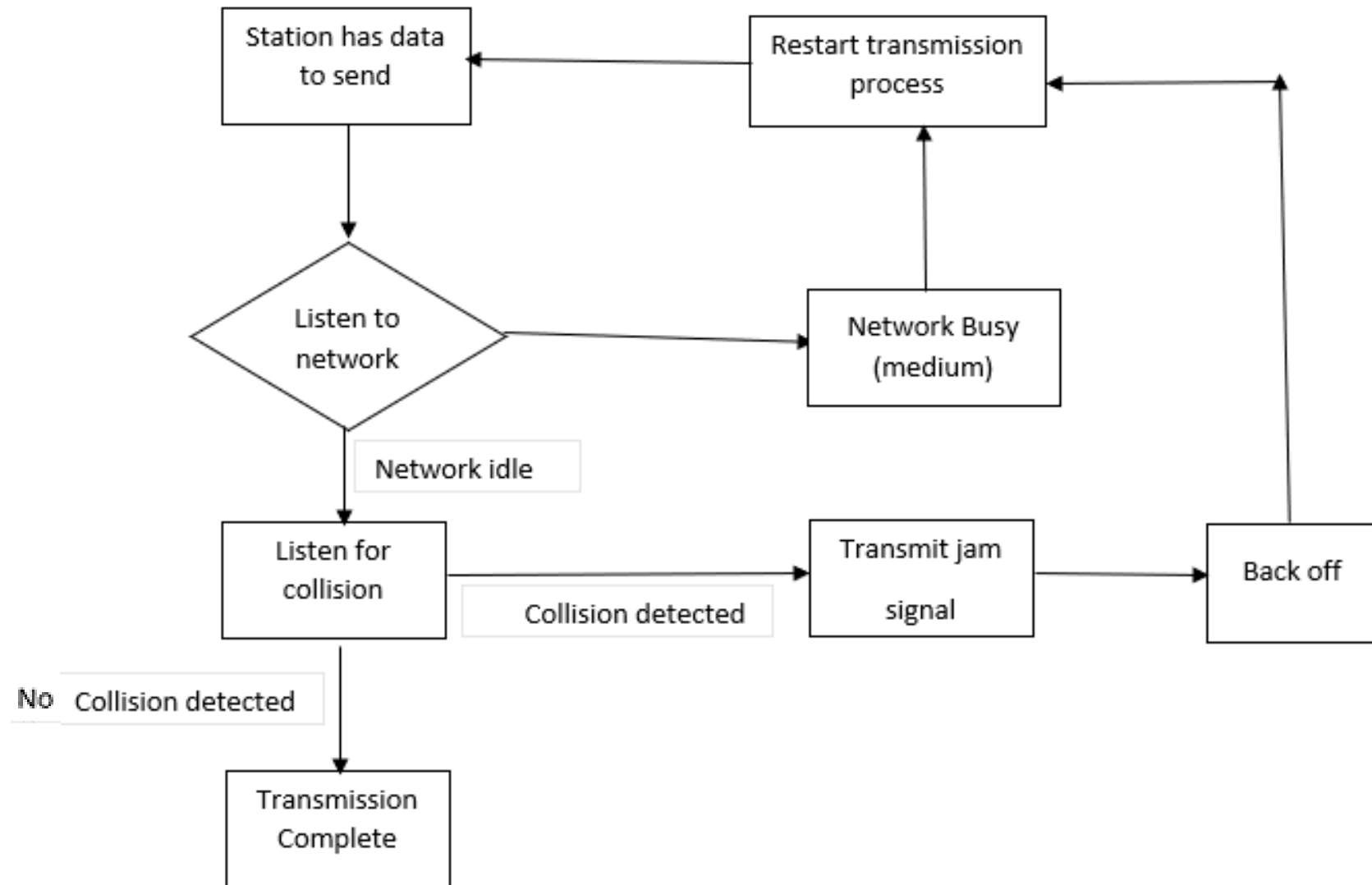
CSMA/CD

- The access mechanism used in ethernet is called CSMA/CD i.e. **carrier sense multiple access with collision detection**
- A station wishing to transmit first listens to the medium to determine if another transmission is in progress (carrier sense)
- If the medium is in use, the station must wait
- If the medium is idle, the station may transmit

CSMA/CD

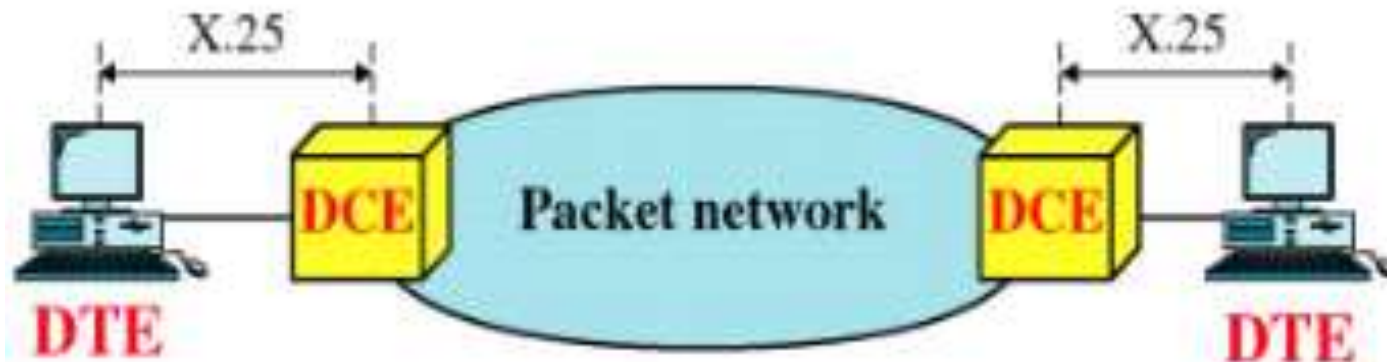
- It may happen that two or more stations attempt to transmit at about the same time
- If this happens, there will be a collision; the data from both transmissions will be garbled and is not received successfully.
 - To account for this, the station waits a reasonable amount of time after transmitting and acknowledgement
 - If there is no acknowledgement, the station assumes that a collision has occurred and retransmits.

CSMA/CD



X.25

- It is an interface between public packet switched networks and their customers
- The figure below gives the conceptual overview of X.25 network
- DTE = Data Terminal Equipment
- DCE = Data Communication Equipment

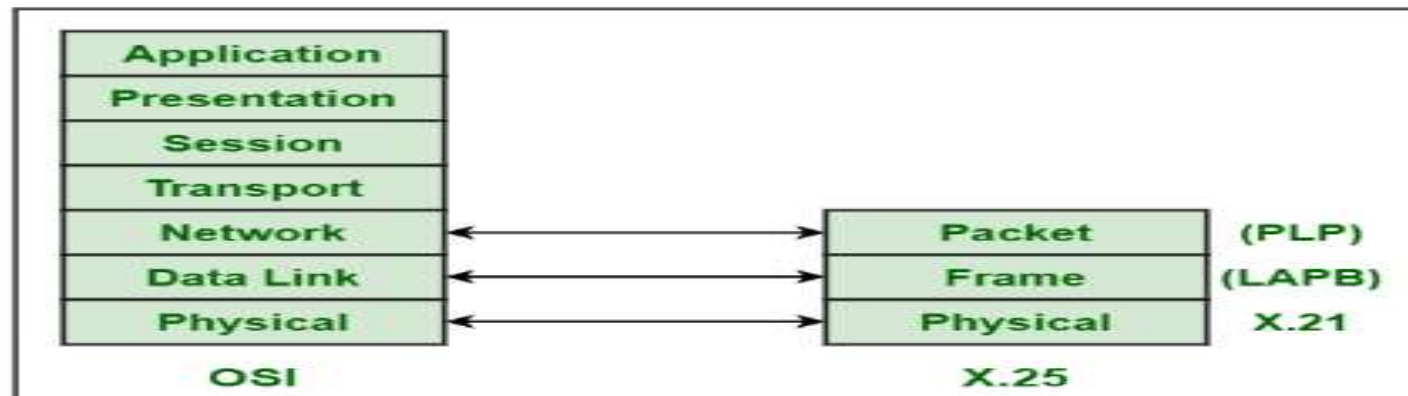


X.25

- It defines how a packet mode terminal can be connected to the other packet network for the exchange of data. It describes the procedures necessary for establishing, maintaining and terminating connections.
- It explains how the user's DTE communicates with the X.25 network and how the packets are sent over that network using DCEs
- X.25 protocol specifies three layers which defines functions at the physical, datalink and network layers of the OSI model
- In 1920's it was largely replaced by a new kind of technology called **Frame Relay**.

X.25

- In 1920's it was largely replaced by a new kind of technology called **Frame Relay**.
- Moreover, if X.25 network is connected to internet, then we know internet already has its own network layer. Hereby, packet layer of X.25 network which is analogous to network layers of internet results ambiguity in function occurring.



X.25 Layer Mapping with OSI Model

Disadvantages of X.25 network

1. Low 64 Kbps rate
 - Most X.25 networks work at speeds upto 64 Kbps, which makes them obsolete from many purposes.
2. It has extensive flow and error control at both data link and network layer. In fact this will create large overhead and slow down the transmission
3. X.25 has its own network layer, so if it is used for the purpose of internet connection, it results to increase in overhead because of duplication of network layers.

Frame Relay

- Frame relay is a relay that connects different LANs.

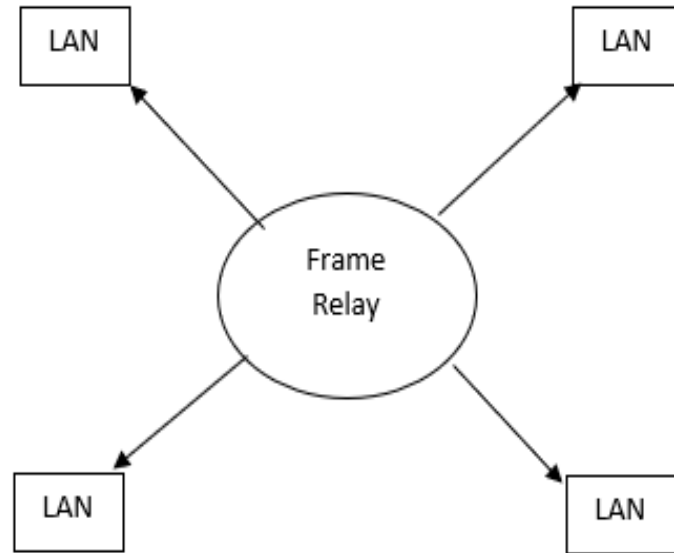


Figure shows interconnection of LANs using frame relay

- Frame relay operates at only the physical and data link layers. Here all error checking is left to the protocols at the network and transport layers which uses the services of frame relay

Advantages - Frame Relay

- It operates at higher speed (1.544 to 44.376 kbps)
- Frame relay operates at only the physical and datalink layers, so there is no duplication if TCP/IP protocol already has a network layer protocol. But for X.25 network, there is duplication of network layer function because X.25 has its own network layer.
- It allows bursty data. It allows a frame size of 9000 bytes, which can accommodate all local area network frames
- It is less expensive than other traditional LANs.
- It can be used as a low cost, high speed backbone for WAN network to connect LAN that do not need real time communication but may have bursty data.

Disadvantages - Frame Relay

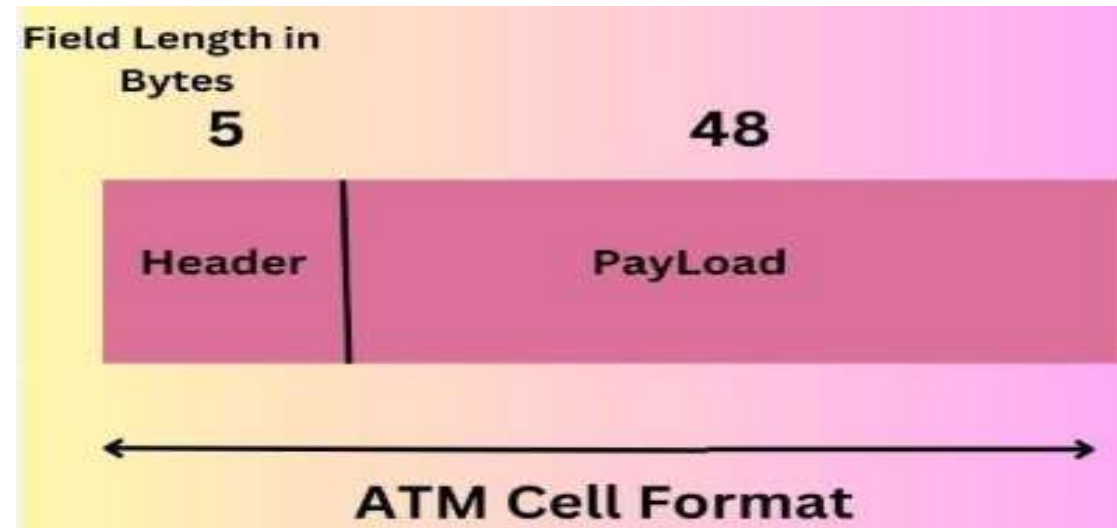
- The data rate of frame relay is still not high enough for protocols with higher data rates (such as BISDN)
- Frame relay allows variable length frames. This may create varying delays for different users.
- Due to varying delays, it is not suitable for sending delay sensitive data such as real-time voice or video. e.g. It is not suitable for teleconferencing.

ATM

- ATM (Asynchronous Transfer Mode)
- ATM is not a frame relay, it is a cell relay
- ATM is a cell relay protocol designed by the ATM forum and adopted by the ITU-T; ATM can be thought of as the “highway” of information superhighway.
- In fact Atm is going to solve the world’s networking and telecommunication problems by merging voice, data, cable television, telex. Telegraph etc. connected by strings into a single integrated system that could do everything for every one

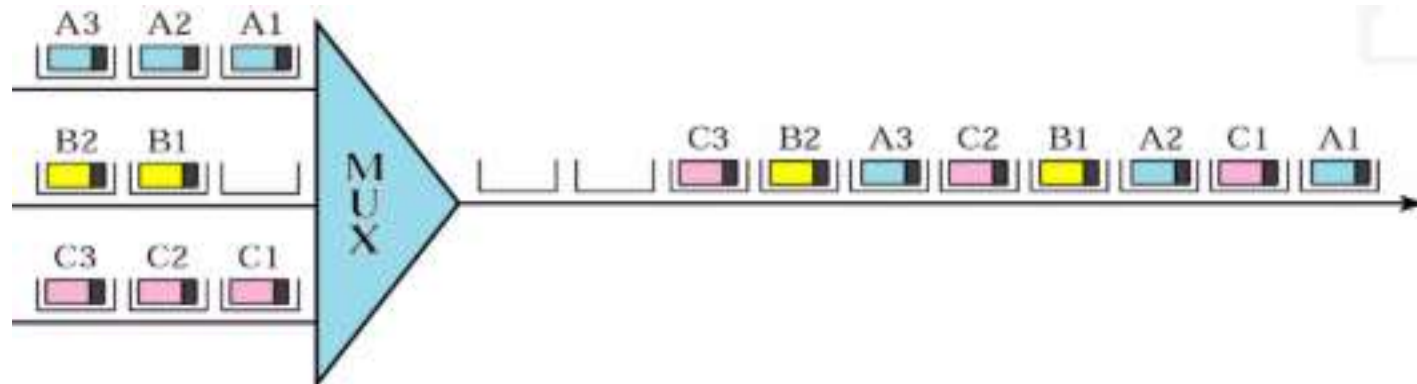
ATM

- The basic idea behind ATM is to transmit all the information in small fixed size packets (or small data size packets) called cells. The cells are 53 bytes long of which 5 bytes are header and 48 bytes are payload (information)



ATM

- Cell routing is done in hardware at high speed.
- Because of the fixed cell size, it is easy to build hardware routers to handle short, fixed length cells.
- In ATM, hardware can be set up to copy one incoming cell to multiple output lines which is required to broadcast television program to many receivers.



ATM

- Small cell does not block any line for very long, which makes guaranteeing quality of service.
- ATM networks are organized like traditional WANs with lines and switches. The most common speed for ATM networks are 155 Mbps and 622 Mbps although higher speeds are also available.

Tutorial

Computer Network

Deadline of submission:

- 1) What are the various services provided by the datalink layer? Explain.
- 2) What is PDU? Differentiate between LLC and MAC sub layer of datalink layer of OSI reference model.
- 3) Explain the operation of CSMA/CD along with the flowchart.
- 4) What is error control? Explain CRC as an error detection mechanism.
- 5) Discuss about sliding window protocol. How datalink layer provides flow control mechanism. Explain giving examples.
- 6) Explain about ALOHA and its types in detail.
- 7) What is the significance of Hamming code in error correction? Explain with a suitable example.
- 8) Compare and contrast stop and wait, go-back-N, selective-repeat request.
- 9) Write short notes on.
 - a. Token ring
 - b. Frame Relay
 - c. ATM
 - d. X.25